

Ethiopian Health Professionals' Licensure Examination

Medical Laboratory Science Exam

Tahisas 2014/December 2021

PART TWO

BOOKLET CODE: 022

Number of Items: 100

SUBJECT CODE: 12

Time Allowed: 2:30 hours

DIRECTIONS: Each of the following questions is followed by four possible alternative answers. Read each question carefully and **blacken** the letter of your best choice on the separate answer sheet provided.

1. A laboratory technologist who is working in a microbiology department was disinfecting laboratory equipments by using highly active compounds against gram positive and gram negative bacteria and viruses including HIV and hepatitis B virus. At the same time, he uses this highly active compound to disinfect discarding containers and for treating spillages of blood.

What is the most likely disinfectant used by the laboratory technologist?

(A) Phenolics

(B) Alcohols

(C) Peroxygen compounds

(D) Chlorine-releasing disinfectants

2. A health center laboratory uses a piece of paper and carton for the collection of stool specimen from the patients. After receiving the specimen, the technologist lets the patients drop the leftover specimen into the toilet. However, most of the patients dropped the specimen outside the toilet.

What is the most appropriate safety precaution should the technologist follow?

(A) Using disposable sample container

(B) Giving disinfectant for the patients

(C) Posting the instruction on the wall

(D) Providing match stick box

3. A laboratory technologist who is working in a research laboratory separates DNA from bacterial samples. Then, he further treats the extracted nucleic acids by phenol solution as part of purification processes.

What function of phenol is considered by the technologist for the process?

(A) Removes polysaccharides from the extracted nucleic acids

(B) Removes RNA from the extracted nucleic acids

(C) Removes lipids from the extracted nucleic acids

(D) Removes proteins from extracted nucleic acids

9. A 15-year-old boy presented to the clinician with sign and symptoms of dull gray, circular patches of alopecia, scaling and itching on the scalp.

What is the appropriate sample collection strategy for the patient?

- (A) Scraping with a toothbrush
(B) Swabbing with cotton tip
(C) Cutting the infected hair
(D) Plucking the infected hair

10. A 64-year-old man develops a small, painless, subcutaneous nodule on his foot that increases slowly but progressively in size. Microscopic examination revealed, dark colored branching intertwining tubes with separation in between and gram-positive partially branching beaded filaments.

What is the most likely pathogen detected from the above patients?

- (A) Bacteria
(B) Virus
(C) Fungus
(D) Parasite

11. A 26-year-old man patient presented to an emergency OPD of a hospital with onset of primary pulmonary infections followed by chronic granulomatous lesions on different parts of the body. The laboratory microscopic result from sputum and biopsies of granulomatous lesions revealed numerous yeast cells with multiple buds attached to the mother cell by very narrow necks.

What will be the causative agent of the disease?

- (A) *Coccidioides immitis*
(B) *Histoplasma capsulatum*
(C) *Paracoccidioides brasiliensis*
(D) *Blastomyces dermatitidis*

12. In a community, 23 people died with acute and chronic hepatitis and liver failure following consumption of improperly stored cereal grains.

What will be the most likely cause of the death?

- (A) Ochratoxin
(B) Aflatoxin
(C) Tetanus toxin
(D) Citreoviridin

13. A 47-year-old farmer came to a health centre with a complaint of abdominal pain, severe itching at his lower leg, and mild pneumonia with cough, sore throat, bloody sputum, headache, weakness and bloody diarrhoea. The clinician ordered a stool examination and the laboratory technologist received the fresh specimen and examined immediately. He found one of the morphological stages of Hookworm under the microscope.

What is the most probable morphological stage of this parasite?

- (A) Filariform larvae
(B) Rhabditiform larvae
(C) Egg stage
(D) Adult stage

24. A 47-year-old patient with known amoebic colitis came to a hospital with complaints of diarrhea and lower abdominal pain. On stool examination, the laboratory technologist processed a direct saline wet mount examination of the fresh stool specimen and found a motile vegetative stage of amoeba which ingested red blood cells.

What is the most likely trophozoite stage of the protozoan?

- (A) *Entamoeba dispar*
- (B) *Entamoeba coli*
- (C) *Entamoeba histolytica*
- (D) *Entamoeba histolytica/dispar*

25. A 20-year-old patient came to a health center with a complaint of abdominal pain. On stool examination, the laboratory technologist observed a large (about the size of two RBCs) pear shaped motile flagellate with a falling leaf motility under the microscope.

What is the most likely parasite and its morphological stage of the observation?

- (A) Oocyst stage of *Cryptosporidium parvum*
- (B) Trophozoite stage of *Giardia lamblia*
- (C) Cyst stage of *Trichomonas hominis*
- (D) Sporozoite stage of *Isospora belli*

26. A 65-year-old patient came to a health center with complaints of abdominal pain and constipation. He also has formed stool with offensive smell. On stool examination, small oval organisms with four nuclei grouped at one end and remnants of flagella were observed under the microscope after Iodine stain.

What is the most likely parasite and its stage found in this patient?

- (A) Trophozoite stage of *Trichomonas vaginalis*
- (B) Cyst stage of *Entamoeba histolytica*
- (C) Cyst stage of *Giardia lamblia*
- (D) Ova of *Ascaris lumbricoides*

27. A 40-year-old patient with ulcerations throughout his body came to a hospital. On laboratory investigation, the laboratory technologist prepared a skin smear from the lesion and stained it using Giemsa stain. He found small, round to oval bodies seen in groups inside monocytes under an oil immersion microscope. The nucleus of these bodies and rod shaped kinetoplasts were dark reddish-mauve stained, while the cytoplasm stained palely.

What was the most likely morphological stage of this parasite?

- (A) Amastigote stage of leishmania species
- (B) Promastigote stage of leishmania species
- (C) Epimastigote stage of leishmania species
- (D) Trypomastigote stage of leishmania species

28. A 50-year-old patient with a history of red papule which looks like fly's bite came to a hospital. On examination, an itchy, enlarged and ulcerated lesion with raised borders, which became hard and crusted with exudating thin serous material from the lesion. On a skin smear examination, the laboratory technologist processed the smear and stained using Giemsa stain and he observed a small, round organism with red staining nucleus and kinetoplast and pale blue staining cytoplasm, lying inside the macrophages and few outside the cell.

What is the most likely way of reporting in this finding?

- (A) Amastigote stage leishmania donovani present
- (B) Amastigotes of leishmania species positive
- (C) Amastigotes of leishmania species present
- (D) Amastigote stage of leishmania seen

29. A 45-year-old HIV/AIDS patient came to a health center with a watery diarrhea. He has high temperature, persistent watery diarrhea, headache, abdominal cramps, nausea and vomiting. On stool examination, a large round mature organism containing two sporocysts each containing two sporozoites under the microscope was found.

What is the most likely parasite and its morphological stage?

- (A) Oocyst of Isospora species
- (B) Oocyst of Sarcocystis species
- (C) Oocyst of Cyclospora species
- (D) Oocyst of Cryptosporidium species

30. A 30-year-old male patient from a rural area arrived at an emergency department with a sudden onset of fever. Upon arrival, his temperature was 39°C and he reported feeling a need to defecate almost constantly. He also has watery diarrhea, headache, abdominal cramps, nausea and vomiting. On stool examination, a small round mature oocyst containing four sporozoites was found under the microscope.

What organism is most likely found in the stool specimen?

- (A) *Isospora belli*
- (B) *Cryptosporidium parvum*
- (C) *Cyclospora cayentanensis*
- (D) *Enterocytozoon bieneusi*

31. A 40-year-old known HIV/AIDS patient was brought to a hospital with complaints of encephalitis with fever, headache, mental deterioration and seizures. After clinical examination, the physician suspected toxoplasma encephalitis and drew a cerebrospinal fluid then sent it to the laboratory for detection of the parasite. The laboratory technologist made a smear of the CSF and stained it using Giemsa stain. He found small, crescent shaped organism (pointed at one end and blunt on the other end) with blue staining cytoplasm and dark red staining nucleus.

What is the most likely morphological stage found in the patient?

- (A) Gametocyte
- (B) Bradyzoite
- (C) Tachyzoite
- (D) Oocyst

32. A 70-year-old patient with complaints of high grade fever, chills, shivering, and arthralgia was sent to the laboratory for blood film examination. The laboratory technologist took a blood sample and found trophozoite stage of *P. vivax*. However, he wanted to see the effect of the parasite on the red blood cells.

What is the most likely method used for the examination of the blood cells?

- (A) Examination of RDT
(B) Examination of thin blood film
(C) Examination of thick blood film
(D) Examination of Fluorescent microscopy

33. A 25-year-old patient, living in a malaria endemic area, was sent to a laboratory for rapid diagnostic test of malaria since the microscope was non-functional. The laboratory technologist observed a positive test result on HRP2 and lactate dehydrogenase lines.

What parasite is the most likely diagnosed in the test?

- (A) *Plasmodium vivax*
(B) *Plasmodium ovale*
(C) *Plasmodium malariae*
(D) *Plasmodium falciparum*

34. A 50-year-old febrile patient was sent to a laboratory for blood film examination. The laboratory technologist examined 100x oil immersion fields using a high power objective and found 80 trophozoite stage of *Plasmodium vivax*.

What is the most appropriate way to report this finding using a plus sign reporting system?

- (A) +1
(B) +2
(C) +3
(D) +4

35. Blood specimen is received at serology laboratory from a patient suspected of jaundice for hepatitis serology. Serologic assays demonstrated the following result.

S/no	Serological Marker	Result
1	HBsAg	Positive
2	Anti-HBs	Negative
3	Anti-HBc	Positive
4	Anti-HBe	Negative

What is the most likely interpretation of this result?

- (A) Immune due to infection
(B) Immune due to vaccination
(C) Acute infection
(D) Chronic infection

A chronic liver disease (CLD) patient came to a hospital for regular follow up. Hepatitis B serology panel was done and the result was sent to the requesting clinician. The clinician advised the patient to be more careful not to transmit to others as he was highly infectious at this stage due to significant viral activity.

Which serological marker showed increased infectivity of the patient?

- (A) The presence of hepatitis B virus DNA
- (B) The presence of anti-hepatitis B core antigen (HBcAg)
- (C) The presence of hepatitis B envelope antigen (HBeAg)
- (D) The presence of anti-hepatitis B surface antigen (HBsAg)

37. A one-year-old baby girl was brought to an emergency department by her parents after two days of persistent watery diarrhea and vomiting accompanied by a low-grade fever. The clinician suspected rotavirus gastroenteritis and sends the patient to the laboratory. The laboratory technologist wanted to perform rotavirus stool antigen test.

Which method is recommended for this particular test?

- (A) Enzyme-linked immunosorbent assays (ELISA)
- (B) Real time PCR (RT-PCR)
- (C) Electron microscopy
- (D) Light microscopy

38. A 37-year-old woman is admitted with malaise, abdominal discomfort and jaundice. The inpatient laboratory collects blood for hepatitis serology and blood chemistry. Blood chemistry showed increased levels of bilirubin and AST. Serologic assays demonstrated the presence of immunoglobulin G antibody (IgG anti-HCV).

What can be concluded from this patient's result?

- (A) Present or past infection
- (B) Ongoing HCV replication
- (C) Confirmed hepatitis
- (D) HCV immunization

39. A one-year-old baby was brought to clinic with complaints of fever, non-productive cough and dyspnea. The clinician suspected bronchiolitis and pneumonia due to viral etiology. The technologist performed virus isolation from respiratory secretion and the findings were enveloped, ssRNA genome and multi nucleated giant cell.

What is the most likely etiological agent in this case?

- (A) Respiratory Syncytial Virus
- (B) Influenza virus
- (C) Measles virus
- (D) Mumps virus

40. A clinician wants to diagnose a neonate born from an HIV-infected mother. The birth attending clinician collects blood sample and send it to a laboratory. The laboratory technologist performed rapid serological (antibody) test and the result was positive.

What is the most appropriate next step in the diagnosis of this infant?

- (A) Repeating the test by another serologic test
- (B) Collecting another sample for RT-PCR
- (C) Reporting the result as indeterminate
- (D) Reporting the result as positive

41. A laboratory technologist, who is working in a health centre, had a discussion on the yearly plan of the health centre with individuals assigned at different levels of the health centre. The initiative was to create a sense of ownership among employees at all levels before official implementation of the plan.

What step of planning is best explained in the above case?

- (A) Establishing objectives for the planning
- (B) Communicating plans with stakeholders
- (C) Situation audit or environmental assessment
- (D) Involving management and staffs for the planning

42. A *woreda* health office conducted a pre-assessment on the existing situation in order to introduce community level health insurance. According to their assessment, the community members were pro health insurance. In addition, gate keepers in the community were positive towards community level health insurance.

What strategic planning tool is best explained in the above scenario?

- (A) Threat
- (B) Strength
- (C) Weakness
- (D) Opportunity

43. A female executive officer's leadership style is full of orders and commands. She expects employees to obey organizational rules and regulations. She also does not give any room for employees' suggestions. As a result, grievance is a common feature of the organization.

What type of leadership style is best expressed in the above case?

- (A) Democratic
- (B) Laissez-faire
- (C) Authoritarian
- (D) Semi-democratic

44. In Ethiopia, the National Health Accounts (NHA) showed that the total spending on health has been growing steadily. In order to have adequate budget for treatment of Malaria the community enrolled in to health insurance and the government also collect money for each service out of one's pocket.

What source of health care financing used in the above scenario?

- (A) Aid and loan
- (B) Privet financing
- (C) Government financing
- (D) Annual budgeting

45. A director of a health centre wanted to assess the reasons for low skilled delivery performance. And the management has decided to use appropriate steps of health management information system to identify the problem.

What is the most likely first step he has to use to know the reason?

- (A) Data collection
- (B) Data processing
- (C) Increasing number of staffs
- (D) Increasing number of rooms

46. The head of the regional laboratory had an intention to get amendments on the ongoing program pertaining to the opening of gene expert at primary hospital level. In order to understand the progress and make a decision, the leader had gathered information via retreat from different stakeholders and finally he is clear with what should be done and how it will be done.

What type of program effectiveness assessment did the head undertake?

- (A) Control
- (B) Monitoring
- (C) Evaluation
- (D) Implementation

47. A pharmaceutical company in one country is choosing a governing approach to its organization from the principle of management. The company has decided to practice for information exchange based on an agreed plan both up and down to the hierarchy of an organization.

What management is the company practicing?

- (A) Division of work
- (B) Unity of direction
- (C) Unity of command
- (D) Centralization of power

48. A leader of the regional laboratory is blamed by his subordinates for being passive and non-directive leader. Moreover, the leader leaves virtually all controls and decision making roles to the group and provides little or no direction, guidance, or encouragement.

What is the type of leadership style that the leader is exercising?

- (A) Democratic
- (B) Laissez-faire
- (C) Authoritarian
- (D) Semi- authoritarian

49. A 35-year-old woman was brought to a laboratory reception unit to give blood sample for pre-operative assessment. The woman was not fully conscious when she reached in the sample collection site. Then, the laboratory technician needs permission or consent of the patient before drawing blood sample.

What type of consent he has to use to draw blood sample based on this scenario?

- (A) Patient autonomy
(B) Informed consent
(C) No need of consent
(D) Surrogate decision maker

50. A laboratory manger needs to distribute packs of gloves for different units of laboratory. The glove was not enough to distribute according to their need. The manger studied the consumption and future need and decided who deserves more of resources to realize the best outcomes of laboratory units.

Which principle of ethics is the manger most concerned?

- (A) Justice
(B) Autonomy
(C) Beneficence
(D) Non-maleficance

51. A 25-year-old pregnant woman visited an ANC clinic. The laboratory professional weighed the mental stability and consciousness of the woman to give blood sample and decided that she can give consent by herself regardless of any other representative.

What type of consent is appropriate to take for the above case?

- (A) Assent
(B) Verbal consent
(C) Written consent
(D) Surrogate decision making

52. A phlebotomist was collecting a blood sample with his colleagues. During the collection process, he was not changing the glove for the new patient unless he believes the glove was torn. One of the clients requested the laboratory worker to use new glove for her, but he is not willing to change the new glove due to different reason.

What is the appropriate recommendation to be followed by phlebotomist in using of gloves?

- (A) Using single glove for single patient for safe practice
(B) Using single glove for multiple patients unless it is torn
(C) Not applying glove for mutual trust of the patient and lab worker
(D) Using single glove for multiple patients if there was shortage of glove

53. A physician examined one of his patients in outpatient department of a rural health center and requested laboratory investigation. When the patient came to the laboratory unit, he had no adequate information and not had good communication between the patient and the laboratory technologist due to long distance between, and a lot of sounds around.

What should the laboratory technologist do to alleviate similar problem permanently?

- (A) Speaking loudly to hear each other
- (B) Just take the request and collect the sample
- (C) Arranging the room as it supports to communicate easily
- (D) Assign another person between to explain what the patient says

54. A senior laboratory technologist is working in one of the regional laboratories of the country. Based on professional standards, he was responsible to serve as laboratory manager, a faculty member in the clinical laboratory education programs, conduct research and consult in the area of laboratory profession.

Which type of education requirement is needed to perform all these responsibilities?

- ☒ (A) Laboratory certificate
- (B) laboratory technician
- (C) Laboratory specialist
- (D) Laboratory technologist

55. A fresh graduate laboratory technologist needs to apply to a vacancy announcement of a hospital human resource as a junior laboratory technologist. When he tried to apply with his CV, temporary degree, and student copy, the human resource officer asked for another document.

What was the most probable requirement expected to be fulfilled?

- ☒ (A) License
- (B) Accreditation
- (C) Certification
- (D) Proficiency result

56. A laboratory professional has been serving in a health center for the last five years. The manger was concerned more about the laboratory's waste disposal system and closely discusses with sanitary professionals about the impact of uncontrolled waste segregation and disposal system to the surroundings.

What type duty is the manager most concerned about based on the above case?

- (A) Duty to the society
- (B) Duty to the profession
- (C) Duty to the colleagues
- (D) Duty to the environment

57. A new graduate laboratory technologist was working in a blood collection unit of a laboratory. Early in the morning, he cleaned the laboratory room and collected all the necessary materials. During blood collection procedure, he never forgot applying glove and using all necessary safety measures. Even though he did not change the glove for each new patient believing it is safe for him.

What most important safety measure was missed by this laboratory technologist?

- (A) Safety for self
- (B) Safety for the patient
- (C) Safety for the colleagues
- (D) Safety for the environment

58. A researcher proposed an experimental study to identify the effect of alcohol on the development of liver disease. However, the ethical review board decided that it is unethical to use experimental study in such scenario and advised the researcher to use another study design.

What study design is best as a substitute to the proposed study design in this scenario?

- (A) Cohort study design
- (B) Case control study design
- (C) Cross sectional study design
- (D) Quasi experimental study design

59. A laboratory technologist, who is a researcher, wanted to test the hypothesis saying the mean blood sugar level is different from 80 mg/dl in population and obtained calculated a Z-statistic value of 1.6 on his analysis. For his decision, he considered the 95% confidence interval reliability coefficient value of $Z=1.96$ as a cut of point for his decision.

What will be his decision?

- (A) Reject the null hypothesis because it is in the non rejection region
- (B) Do not reject the null hypothesis since z- statistic is less than 1.96
- (C) Do not reject the null hypothesis because it lies in the rejection region
- (D) Reject the null hypothesis since the probability of obtaining z- statistic by chance is less than 1.96

60. A laboratory technologist, working in district hospital, wanted to study the burden of *taenia-saginata* and factors contributing to the infection among sixth grade students in a school.

What are the most appropriate data collection tools for this study?

- (A) Tape recorder and microscope
- (B) Data extraction checklist and microscope
- (C) In-depth interview guide and microscope
- (D) Interviewer administered questions and microscope

61. A researcher obtained an informed consent from an HIV positive study participant for the purpose of estimating the prevalence of the disease in a community. The researcher used the same data to identify list of HIV positive individuals in the community for another study without asking their consent again.

What ethical principle is breached?

- (A) Justice
(B) Autonomy
(C) Beneficence
(D) Non-maleficence

62. A laboratory technologist is working in central statistic authority as data analyst. He wanted to estimate the average variation of study participants by age using a right skewed data from a demographic and health survey 2016 (DHS2016).

What is the most appropriate measure of dispersion for these types of data?

- (A) Range
(B) Variance
(C) Standard deviation
(D) Inter quartile range

63. A laboratory technologist who is working in a hospital as researcher conducted a study regarding the effect of bed net use on malaria infection among pregnant mothers using a cohort study. After running the analysis, relative risk 3 was found for no bed net use.

What is the most appropriate interpretation of the researcher's finding?

- (A) Pregnant mothers who do not use bed net have half the risk of having malaria.
(B) The risk of not using bed net is twice among mothers with malaria than those without malaria.
(C) Not using bed net causes malaria with 3 times risk whereas using bed net do not have effect on malaria among pregnant mothers.
(D) The rate of having malaria infection is three times higher among mothers who do not use bed net compared to mothers who use bed net.

64. A researcher assessed the performance of a laboratory screening test and found a sensitivity of 90%.

What is the correct interpretation of the finding?

- (A) The probability that a person tested positive actually have the disease is 90%.
(B) The probability that a person tested negative do not have the disease is 90%.
(C) The probability of the test registers positive result for a person with the disease is 90%.
(D) The probability of the test registers negative result for a person who did not have the disease is 90%.

65. A disease surveillance department of a *woreda* health office received message through telephone call about the occurrence of an outbreak in one of its *kebeles* and decided to conduct investigation. On next day, a surveillance team gets well prepared and ready for the field work.

What is the most appropriate next step to be done in this case?

- (A) Verify the diagnose
- (B) Develop hypotheses
- (C) Construct a working case definition
- (D) Establish the existence of an outbreak

66. There exists an immediately reportable disease in a *woreda*. A surveillance department of the *woreda* health office wanted to run sentinel surveillance on the disease.

What is the most appropriate approach to be followed to conduct the surveillance?

- (A) Receive routine report that come to the *woreda* from a health facilities
- (B) Initiate and collect a report regularly from health facilities in the *woreda*
- (C) Select sample of health facilities and receive report regularly from them
- (D) Wait being alert for any kind of report on the disease that may come to the department

67. A local health extension worker wants to conduct a health education on child feeding to mothers in her *kebeles*. In order to make her messages more retained, she wanted to use a combination of teaching materials.

What is the most appropriate combination of health promotion materials in this case?

- (A) Audio and leaflet
- (B) Video and demonstration
- (C) Health talks and mass media
- (D) Health talks and photographs

68. A hospital implemented the health intervention measures which are intended to reduce impairments, disability and suffering caused by health problems. The hospital also wants to promote patients' adjustment to not curable conditions.

What level of diseases prevention will be considered for this intervention?

- (A) Primordial prevention
- (B) Primary prevention
- (C) Secondary prevention
- (D) Tertiary prevention

69. Hand washing as a primary method of infection prevention has been a cornerstone of infection control program since the infection control profession began in the early 1970s. Health professionals at all levels gave high concern for hand washing practice.

What is the recommended product for hand washing practice?

- (A) Waterless, alcohol-based hand rub are now the recommended product
- (B) Alcohol-base with water hand rub is now the recommended product
- (C) Antimicrobial-impregnated wipes are recommended product
- (D) Plain soap and water are now the recommended product

70. A health professional who is working in a hospital on the process of providing healthcare in any setting may acquire infections. A laboratory technologist was delegated as an infection prevention focal person in the hospital.

Which one is the most appropriate practice to prevent infection in the work environment?

- (A) Identify work-related infection risks and institute appropriate preventive measures
- (B) Identify work-related surface used for the appropriate preventive measures
- (C) Identify work-related nutrition used to prevent the occurrence of infection
- (D) Identify work-related mask used in order to prevent infection

71. A laboratory technologist was assigned in a hospital received a contaminated cultures as a health care waste which needs appropriate handling and treatment.

What is the most appropriate treatment?

- (A) Burning in the incinerator
- (B) Burning in the pit
- (C) Sterilizing by autoclave
- (D) Sterilizing by incinerator

72. A laboratory technologist was assigned as a team member of epidemic investigation, a number of circumstances may lead to an outbreak or epidemic in the community. In order to implement intervention to prevent and control an outbreak the team needs to identify the cause of outbreak.

Which situations is the most likely cause of outbreak?

- (A) The result of human inappropriate interaction with the environment and social relationship to secure human needs.
- (B) The result of human miss management of diseases etiology as a result of social interaction and the environment.
- (C) The result of some ecologic imbalance among the host, the agent, and the environment.
- (D) The result of human exposure without considering the environmental situations and his/her resistance.

73. A laboratory technologist working in a primary hospital noticed that the number of people being screened for Hepatitis B virus is low despite the high prevalence of Hepatitis B in the region. When he assesses the reason why there is low screening rate, he realized that there is shortage of Hepatitis B screening kit.

What is the most likely behavioral factor that affects the rate of screening?

- (A) Predisposing factor
- (B) Influencing factor
- (C) Enabling factor
- (D) Reinforcing factor

74. A laboratory professional who wanted to initiate a screening test on a disease, considered the test need to be simple and safe that it does not cause harm to the people.

What other criteria does the professional needs to consider?

- (A) The disease must have a reasonably long lead time
- (B) Use a very highly specific test regardless of the sensitivity
- (C) Use a very highly sensitive test regardless of the specificity
- (D) The screening can be initiated regardless of prevalence of the disease

75. In Ethiopia, based on the current community health problems, the experts decide on the need of additional health problems, add to the existing health problems under the National Notifiable Disease Surveillance System (NNDSS) at the Federal Ministry of Health (FMoH) level.

What is the condition of the disease target to NNDSS?

- (A) Reports of cases of specific diseases of public health importance
- (B) The community leader complain about the specific health problem
- (C) Report of repeated patients of undetected cases from the community
- (D) The complain of higher health professionals about specific health problem

76. A laboratory technologist is interested to prepare an action for individuals with in his department. To do so, he is worried about what components should be incorporated in his plan.

What most appropriate component should be incorporated in his plan?

- (A) An organization's mission.
- (B) Strategic directions to the goal.
- (C) Detail objectives of work groups and individuals.
- (D) Depicts an outline of steps designed with the goals of the entire organization

77. A glucose standard having a concentration of 100 mg/dl was purchased by the laboratory. The standard was measured several times to assess the method's performance in the laboratory. All of the results were found to be much lower than the expected value, but each reading had almost the same value.

What can we say about the method's performance?

- (A) The method had good accuracy than precision
- (B) The method was accurate and precise
- (C) The method was accurate
- (D) The method was precise

78. A new diagnostic rapid kit was evaluated using 200 malaria suspected specimens. The laboratory technologist reported 20 samples as positive using the new method. The confirmatory PCR technique, confirmed 15 of the samples as positive.

What was the negative predictive value of the new method?

- (A) 97.3% (B) 99% (C) 100% (D) 75%

79. A laboratory technologist was interpreting LJ chart after recording daily control results of 2 ranges of control materials for 20 consecutive days. During interpretation process, one observation exceeded the mean plus 2SD and another exceeding the mean minus 2SD.

What is the interpretation based on the above case?

- (A) Trend was observed (C) R_{45} rule was violated
(B) Shift was observed (D) 2_{25} rule was violated

80. A team of experienced laboratory professionals from different specialties was organized to develop SOPs. The team reviewed clearly, what the current test protocol requires in different laboratory test procedures and developed a clear and understandable SOP which fulfilled all the requirements of good SOP. The manager printed all the SOPs and put on the shelf at the office and allowed each individual to refer SOPs coming to his office when needed.

What was the most failure of the laboratory manager regarding the above SOP?

- (A) Non-comprehensive SOP preparation (C) Inaccessibility of the SOP
(B) Non-understandable SOP preparation (D) Unclear SOP preparation

81. A laboratory professional was running a newly purchased quality control materials for 20 consecutive days using the standardized method and registered each control result on the Levy Jennings chart. When the laboratory technologist evaluated individual results, 3 readings were found out of the plus 3 standard deviation from two levels of control materials.

What should the laboratory professional do with regard to the above control?

- (A) Re-evaluating the control for another 20 days
(B) Do not use the control for patient sample reporting
(C) Validating the machine to increase the performance
(D) Using the control for monitoring the analytical phase

82. The laboratory technologist was performing a complete blood cell counting (CBC) using a hematology analyzer. When he checks the result, red blood cell count, hemoglobin value and hematocrite value were, 4×10^9 /ml of blood, 20 g/dl, and 30% respectively.

What should the laboratory technologist decide on the above case?

- (A) Reporting the result of RBC count
- (B) Reporting the result of hematocrite value
- (C) Reporting the result of hemoglobin value
- (D) Running control material and doing again

83. A regional laboratory has planned to establish a new laboratory for viral culturing and corona virus vaccine development besides its current diagnostic services. The laboratory manger organized a team to study the laboratory's design and safety requirements during corona virus vaccine development.

What type of laboratory should the team recommend for the given purpose?

- (A) Maximum containment laboratory
- (B) Safety level II laboratory
- (C) Containment laboratory
- (D) The routine basic laboratory

84. A urine sample was sent to a laboratory for parasitological examination and bacteriological culture. During parasitological examination, *Schistosoma mansoni* was found so that the technologist had rejected the urine culture.

What is the possible reason for rejecting the sample?

- (A) Unknown delay of the sample
- (B) Contamination with feces
- (C) Unpreserved urine
- (D) Bloody urine

85. Diarrheal outbreaks were occurred in remote area of a district. The patients had bloody diarrhea and other symptoms of gastrointestinal tract. The laboratory technologist at the field needed to collect and transport the stool specimens to the bacteriology laboratory for bacterial culture.

What is the appropriate medium to transport the specimen?

- (A) Amines medium
- (B) Cary-Blair medium
- (C) Nutrient medium
- (D) Thioglycollate medium

86. A senior supervisor in diagnostic laboratory recognises the presence of systemic error when performing different clinical chemistry tests. He ordered other technologist to prepare a standard solution with different concentration for calibration.

How can the technologists prepare these standard solutions?

- (A) By preparing the standard solution by $C1V1 = C2V2$ dilution
- (B) By preparing the standard solution by proportion dilution
- (C) By preparing the standard solution by simple dilution
- (D) By preparing the standard solution by serial dilution

87. A laboratory technologist who is working at district laboratory has a semi- auto method spectrophotometer at his laboratory. Unfortunately, the service duration of this instrument is too short.

What simple preventive maintenance he can apply to improve shelf life?

- (A) Changing spares daily
- (B) Keeping it always on
- (C) Cleaning it daily
- (D) By not using it

88. A laboratory technologist who is working at a health centre laboratory receives a new clinical chemistry analyzer. Before he starts the installation process, he wants to be sure about the physical requirements.

What are the physical requirements he has to be sure about?

- (A) Space, doors, ventilation, water supply, electric power
- (B) Standard solution, control solution, sample solution
- (C) Reagent, patient specimen, recording log book
- (D) Money, reagent, patient specimen

89. A laboratory technologist at a molecular biology laboratory performed RT-PCR for HIV. While doing the test, he observed greater fraction of primers bind to the incorrect site on the template DNA.

What should the technologist do to avoid this problem?

- (A) Reducing the amount of primer
- (B) Increasing the temperature to 75
- (C) Lowering the reaction temperature to 37
- (D) Using a different type of DNA polymerase

90. A laboratory technologist was assigned to a quality officer position at a molecular laboratory. At the beginning, he tried to identify sources of errors from the occurrence management review logbook.

What is the source of error in the post-analytical phase?

- (A) Specimen labeling error (C) Malfunctioning instrument
(B) Specimen contamination (D) Transcription error

91. A physician has sent a CBC request for a 25-year-old patient. A laboratory technologist collected venous blood and analyzed the specimen using a five part differential haematology analyzer. However, the analyzer displayed flag sign on WBC, RBC and platelet counts so that the technologist repeated the test and the same flag errors were detected. The technologist also checked that the control results and background counts were within the acceptable limits.

What is the most likely next action that should be taken by the technologist?

- (A) Prepare a blood film to verify the automated count
(B) Reject the result and request new blood sample
(C) Re-calibrate the haematology analyzer
(D) Report the result to the physician

92. A laboratory technologist, who is working at a hospital laboratory, has been processing many patient samples per day in an automated haematology analyzer. As a routine practice, he usually reviewed patient results before reporting to the physician. One day, he suspected the reliability of one test result and wanted to check the result before reporting to the physician.

What is the most likely action does the technologist perform?

- (A) Run control samples for verification
(B) Record and report the patient result
(C) Check the procedure for accuracy
(D) Perform delta check

93. The laboratory technician who is working at a hospital laboratory is analyzing lots of patient blood samples daily as a routine procedure in haematology analyzer. While he is running the patient sample, the analyzer shows an error sign. The laboratory technician performs trouble shooting action using the user manual based on the error sign and solves the problem.

What is the most likely next step the technician should do?

- (A) Repeat the patient sample
(B) Inform the problem to the lab manager
(C) Repeat the sample using manual method
(D) Properly document corrective actions taken

94. A regional laboratory inspector wanted to evaluate the quality of Parasitology laboratory performance of the district laboratory as part of onsite supervision. While the district laboratory technologist performing stool examination, he wanted to observe protozoan cysts using iodine wet mount preparation. On his assessment he observed cyst stage of *G. lamblia* under the microscope. However, the cyst stage had green cytoplasm, black glycogen material and red refractile nuclei. The inspector observed the cyst and he told him that this colour reaction is wrong.

What appropriate action should the laboratory technologist have taken?

- (A) Perform the direct saline wet mount examination and report
 - (B) There was no problem with the procedure and the iodine
 - (C) Inspect the iodine solution and discard if it was too light
 - (D) Preserve the stool and send it to reference laboratory
95. A 23-year-old patient diarrheic stool sample was sent to the laboratory for microscopic evaluation. Before processing the stool specimen, the technologist inspected the stool container and there was a leak from the container.

What appropriate measure should the technologist take to alleviate this problem?

- (A) Proceed with examination the stool
 - (B) Take a blood sample instead and report
 - (C) Reject the stool and order another sample
 - (D) Discuss with the patient about the issue and process
96. An instrument for HIV RNA quantification which installed and calibrated initially by the manufacturer was found to be malfunctioning. Though they repeat preventive maintenance and troubleshooting process, the problem was not fixed. The quality officer decided the instrument should be recalibrated as they start new reagent lot.

How can he recalibrate the instrument?

- (A) By checking the sample quality with its storage condition
- (B) By sending the machine it to the company for recalibration
- (C) By running known negative sample which is prepared in the lab
- (D) By comparing the new reagent lot with previously known reagent lot

97. A quality officer gives training for the newly recruited medical laboratory personnel on what should be the content of report of the patient result look likes by using the existing standard according to ISO 15189.

What should be the most optional information in the content?

- (A) The identification of the laboratory doing the test
- (B) Patient identification and patient location
- (C) Identification of all examinations done
- (D) Date of primary sample collection

98. A respiratory specimen was collected from rural community to isolate the suspected viral agents. Before the specimens have been transported to the regional laboratories, the laboratory manager oriented the specimen collectors to be careful and use viral transport medium as well as cold chain transportation at 4°C.

Which method is most likely needs the specified type of specimen transportation?

- (A) Molecular methods
- (B) Antigen detection
- (C) Antibody detection
- (D) Virus isolation

99. A newly graduated laboratory technologist has joined a tertiary care hospital. He has been assigned to the serology laboratory. He always processes blood, serum and plasma specimens for serological tests.

What are the most likely pathogens he may acquire while processing and testing?

- (A) HBV, HCV, HIV
- (B) HAV, HEV, Polio
- (C) Influenza, measles, mumps
- (D) HAV, HBV, HCV

100. A reference laboratory sent proficiency test (PT) samples to different laboratories having the same set-ups. The laboratories analyzed and sent back the result to the reference laboratory. The reference laboratory, analyzed the results and decided on the overall performance of the laboratories participated in the PT program.

Which part of the laboratories' phase is best addressed in this case?

- (A) Overall performance of the laboratory
- (B) Post analytical phase of testing
- (C) Pre-analytical phase of testing
- (D) Analytical phase of testing

THE END